RESILIENCE REVIEW

Multi-region: the cost of a second AWS region

What it would take to survive the loss of a whole AWS region: run cost, build cost and time, people, and where to put it.

1 service has no multi-region profile (Transform): costed by their catalog category, and their availability in other regions is not verified.

SCOPE
name contains 'prod'

ACCOUNTS GOING MULTI-REGION
40 of 40

PRIMARY REGION
Europe (London) (eu-west-2)

SECOND REGION
Europe (Frankfurt) (eu-central-1) · recommended

RECOVERY TARGET
Recover within 3 hours of a region-wide event

DATA RESIDENCY
Anywhere AWS has a region

1 Executive summary

BOTTOM LINE

To recover within 3 hours, a second region in Europe (Frankfurt) for 40 accounts would add roughly $2.5m to $6m a year to run (25 to 70% of their spend), cost $1.6m to $3.5m and take 5 to 11 months to build, and need 2 to 5 more people.

> **Figures are provisional.** They use default answers. Choose the recovery target and accounts, and answer the questions in section 2.

WHAT WE FOUND

- **Today.** 90% of regional spend runs in Europe (London); Europe (Ireland) already carries some.
- **Recommended region.** Europe (Frankfurt): 100% of spend is on services available there, about 15 ms from Europe (London), prices about +2% against it.
- **Runners-up.** Europe (Ireland): no Connect here ($20.9k/month to re-architect). US East (N. Virginia): ~80 ms further than Europe (Frankfurt).
- **Failover risk.** 5 of 40 chosen accounts run a material service that cannot meet the target as built (no live replica: restore takes hours).
- **Data residency.** AWS has no second region in the United Kingdom: any second region moves data across a border, which needs regulator and data-protection sign-off (UK GDPR; PRA SS2/21).
- **People.** About 2 to 5 extra FTE to run two regions ($400k to $800k a year).

WHAT WE RECOMMEND

1. **Settle residency first.** Agree with the regulator which data may sit outside the country.
2. **Tier the workloads.** Not every account needs the same target: keep instant failover for the few systems whose outage is a systemic event, and 3-hour recovery for the rest.
3. **Fix the hardest services first.** Start with services that have no live replica or are missing in the second region; they set the timeline.
4. **Rehearse.** Prove failover with a scheduled region-evacuation test and keep the evidence.

Extra cost to run (a year)

$2.5m to $6m PROVISIONAL

25 to 70% of the chosen accounts' spend

Cost to achieve (one-off)

$1.6m to $3.5m PROVISIONAL

4.5 to 10 person-years

Time to achieve

5 to 11 months PROVISIONAL

bound by team capacity

Extra people

2 to 5 FTE PROVISIONAL

$400k to $800k a year

Recommended region

Europe (Frankfurt)

~15 ms · 100% service parity

Accounts with failover risk

5 of 40

a material service cannot meet the target as built

2 Your choices

Saved in this browser only. Team, rates and delivery answers are shared with the exit estimate.

PROVISIONAL Using default answers. Change any answer, or confirm these.

Recovery target after a region-wide event

Recover within 3 hours

Systemic risk event: data replicated continuously, a minimal footprint scaled up on failover (pilot light).

Where may the second region be? **Anywhere AWS has a...**	Primary region **Detected: Europe (Lo...**	Second region **Recommended: Euro...**	AWS Support plan **Enterprise**
People in the core cloud team **10**	% of their time on this programme **50**	Application engineers available (FTE) **10**	Blended day rate **750**
Currency **USD**	Cost per extra FTE a year (0 = day rate × 220) **0**	Standard CI/CD pipeline? **Several, partially sta...**	Terraform, and is it up to date? **Terraform with signifi...**

3 Accounts to make multi-region

Sorted by spend. Readiness is the worst failover risk among an account's material services for the chosen recovery target; open an account to see each service's risk.

Account	Spend / month	Readiness	Multi-region risks
data-platform-prod 561017993421 · Root / Workloads / Production	$133k	High	▾ High 5 · Moderate 3 · Low 2 **High** **Redshift** $62k/mo · No live replica: restore or re-run No live cross-region replica. Copy snapshots to the second region and restore (hours for a large warehouse), or load both regions from the pipelines. **High** **Glue** $15k/mo · No live replica: restore or re-run Jobs, crawlers and the Data Catalog are regional: replicate the catalog and re-run pipelines in the second region. **High** **EMR** $11k/mo · No live replica: restore or re-run Clusters are rebuilt in the second region from code; data must be replicated to S3 there. **High** **Athena** $6.6k/mo · No live replica: restore or re-run Queries need the data and Glue catalog to exist in the second region. **High** **MWAA (Airflow)** $2.0k/mo · No live replica: restore or re-run Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours. **Moderate** **QuickSight** $5.1k/mo · Manual or scripted failover Has its own failover model, often only for specific region pairs. **Moderate** **EC2 Other (EBS, NAT, transfer)** $3.7k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **Firehose** $2.6k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Low** **S3** $22k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. **Low** **Lake Formation** $640/mo · AWS replicates and fails over Pipelines must deploy to both regions; the tooling itself can stay in one.

Account	Spend / month	Readiness	Multi-region risks
core-banking-prod 813958344723 · Root / Workloads / Production	$122k	Moderate	▼ Moderate 7 · Low 1 **Moderate** **EC2** $56k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $34k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2 Other (EBS, NAT, transfer)** $15k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **FSx** $5.7k/mo · Manual or scripted failover Cross-region backup copy, or SnapMirror for FSx for NetApp ONTAP; restore or break the mirror to fail over. **Moderate** **Direct Connect** $4.6k/mo · Manual or scripted failover Needs connections or a Direct Connect gateway association into the second region; physical lead times are weeks to months. **Moderate** **ELB** $2.0k/mo · Manual or scripted failover Load balancers are regional; Route 53 health checks switch traffic between them. **Moderate** **Amazon MQ** $1.4k/mo · Manual or scripted failover Needs a cross-region replica. Asynchronous replication loses the last seconds of writes, and promoting the replica is a decision someone has to make. **Low** **Backup** $3.4k/mo · AWS replicates and fails over Copy backups to a vault in the second region; restore time grows with data size.

Account	Spend / month	Readiness	Multi-region risks
payments-core-prod 534439589175 · Root / Workloads / Production	$116k	Moderate	▾ Moderate 10 · Low 2 **Moderate** **EC2** $42k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $28k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2 Other (EBS, NAT, transfer)** $9.9k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Moderate** **MSK (Kafka)** $7.9k/mo · Manual or scripted failover MSK Replicator copies topics cross-region; consumers must handle offset translation on failover. **Moderate** **EKS** $4.6k/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Moderate** **ElastiCache** $4.2k/mo · Manual or scripted failover Global Datastore (Redis OSS / Valkey) replicates cross-region; without it the cache starts cold after failover. **Moderate** **CloudWatch** $3.6k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. **Moderate** **ELB** $2.3k/mo · Manual or scripted failover Load balancers are regional; Route 53 health checks switch traffic between them. **Moderate** **Lambda** $1.2k/mo · Manual or scripted failover Functions are regional: deploy to both regions and

Account	Spend / month	Readiness	Multi-region risks
			route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. Moderate **KMS** $978/mo · Manual or scripted failover Keys are regional. Data encrypted with single-Region keys cannot be decrypted in the second region: use multi-Region keys and re-encrypt. Low **S3** $4.5k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. Low **DynamoDB** $2.8k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.

Account	Spend / month	Readiness	Multi-region risks
cards-processing-prod 966773840736 · Root / Workloads / Production	$101k	High	▾ High 1 · Moderate 8 · Low 2

High **Cognito** $2.4k/mo · No live replica: restore or re-run

User pools are regional with no replication and password hashes cannot be exported: users must be migrated or re-registered, or a sync built.

Moderate **Lambda** $23k/mo · Manual or scripted failover

Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.

Moderate **API Gateway** $9.8k/mo · Manual or scripted failover

Deploy the API in both regions behind a custom domain with Route 53 failover.

Moderate **Kinesis Streams** $8.4k/mo · Manual or scripted failover

No native cross-region replication: producers dual-write or a consumer replicates; consumers restart at a new position.

Moderate **Step Functions** $7.1k/mo · Manual or scripted failover

State machines are regional; running executions do not fail over and must be restarted.

Moderate **CloudWatch** $4.8k/mo · Manual or scripted failover

Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.

Moderate **SQS** $3.2k/mo · Manual or scripted failover

Queues are regional and messages are not replicated: in-flight messages are lost or delayed on failover.

Moderate **EventBridge** $1.7k/mo · Manual or scripted failover

Global endpoints can fail event buses over between two regions; rules and targets must exist in both.

Moderate **KMS** $1.4k/mo · Manual or scripted failover

Keys are regional. Data encrypted with single-Region keys cannot be decrypted in the second region: use multi-Region keys and re-encrypt.

Low **DynamoDB** $35k/mo · AWS replicates and fails over

Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-

Account	Spend / month	Readiness	Multi-region risks
			wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. ● Low **S3** $4.3k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
digital-channels-prod 978681853473 · Root / Workloads / Production	$57k	High	▾ High 1 · Moderate 8 · Low 1 **High** **Cognito** $5.6k/mo · No live replica: restore or re-run User pools are regional with no replication and password hashes cannot be exported: users must be migrated or re-registered, or a sync built. **Moderate** **ECS** $10k/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. **Moderate** **RDS / Aurora** $9.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2** $8.0k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **AppSync** $4.3k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Moderate** **Pinpoint** $2.8k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Moderate** **SNS** $1.8k/mo · Manual or scripted failover Topics and subscriptions are regional: recreate in the second region and publish there. **Moderate** **WAF** $1.5k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. **Moderate** **Amplify** $1.3k/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over.

Account	Spend / month	Readiness	Multi-region risks
			Low **S3** $2.5k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
ml-platform-prod 217230843944 · Root / Workloads / Production	$47k	Moderate	▾ Moderate 6 Low 2 Moderate **SageMaker** $19k/mo · Manual or scripted failover Models and endpoints must be redeployed in the second region; training data replicated. Moderate **EC2** $6.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **OpenSearch** $4.7k/mo · Manual or scripted failover Cross-cluster replication to a domain in the second region; re-point clients on failover. Moderate **Textract** $3.0k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. Moderate **Bedrock AgentCore** $1.2k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. Moderate **Transform** $495/mo · Manual or scripted failover No cross-region failover built in. Images, launch templates and capacity must exist in the second region and be started by a runbook or automation. Low **Bedrock** $7.6k/mo · AWS replicates and fails over Cross-region inference spreads requests across regions, but model availability differs: check every model used exists in the second region. Low **S3** $4.2k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
shared-network-prod 263737281768 · Root / Infrastructure	$37k	Moderate	▼ Moderate 4 Moderate **VPC** $12k/mo · Manual or scripted failover VPCs, subnets, endpoints and inter-region peering or Transit Gateway must be built in the second region. Moderate **EC2 Other (EBS, NAT, transfer)** $8.3k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. Moderate **Direct Connect** $6.6k/mo · Manual or scripted failover Needs connections or a Direct Connect gateway association into the second region; physical lead times are weeks to months. Moderate **Network Firewall** $5.3k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.

Account	Spend / month	Readiness	Multi-region risks
contact-centre-prod 125285552493 · Root / Workloads / Production	$34k	Moderate	▾ Moderate 5 · Low 2 **Moderate** **Connect** $21k/mo · Manual or scripted failover Connect Global Resiliency replicates instances and numbers between specific region pairs and is enabled by AWS on request; otherwise a second instance with its own numbers. **Moderate** **Transcribe** $3.4k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. **Moderate** **Lex** $3.1k/mo · Manual or scripted failover Models, endpoints and quotas differ by region and must be provisioned in the second region. **Moderate** **Lambda** $2.3k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **Kinesis Streams** $1.3k/mo · Manual or scripted failover No native cross-region replication: producers dual-write or a consumer replicates; consumers restart at a new position. **Low** **DynamoDB** $1.8k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. **Low** **S3** $1.5k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
fraud-analytics-prod 466806226017 · Root / Workloads / Production	$20k	High	▾ High 1 · Moderate 4 · Low 1 **High** **Managed Flink** $3.6k/mo · No live replica: restore or re-run Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours. **Moderate** **Neptune** $6.8k/mo · Manual or scripted failover Global Database for cross-region replicas; failover is a promotion step. **Moderate** **SageMaker** $4.3k/mo · Manual or scripted failover Models and endpoints must be redeployed in the second region; training data replicated. **Moderate** **Timestream** $2.0k/mo · Manual or scripted failover Needs a cross-region replica. Asynchronous replication loses the last seconds of writes, and promoting the replica is a decision someone has to make. **Moderate** **Lambda** $994/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Low** **DynamoDB** $2.6k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.

Account	Spend / month	Readiness	Multi-region risks
doc-store-prod 454519794043 · Root / Workloads / Production	$9.7k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $4.8k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.9k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $986/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $975/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
id-verify-prod 267387487076 · Root / Workloads / Production	$8.7k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $4.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $877/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $877/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
investments-prod 514740085648 · Root / Workloads / Production	$8.5k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $5.1k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $1.7k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $1.7k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
mobile-backend-prod 434423300025 · Root / Workloads / Production	$8.0k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $2.8k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $2.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $1.6k/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $1.2k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.
identity-prod 189183446249 · Root / Infrastructure	$7.9k	Moderate	▾ Moderate 3 Low 1 Moderate **EC2** $3.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **Directory Service** $2.2k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. Moderate **ACM / Private CA** $1.6k/mo · Manual or scripted failover Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. Low **Secrets Manager** $647/mo · AWS replicates and fails over Turn on secret replication to the second region.

Account	Spend / month	Readiness	Multi-region risks
payroll-api-prod 295250496497 · Root / Workloads / Production	$7.6k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $3.8k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.3k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $758/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $755/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
branch-queue-prod 717640922470 · Root / Workloads / Production	$7.3k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $3.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.2k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $732/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $728/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
insurance-quotes-prod 498633168856 · Root / Workloads / Production	$7.2k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $3.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.2k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $715/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $719/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
reporting-hub-prod 244219422241 · Root / Workloads / Production	$7.1k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $4.3k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $1.4k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $1.4k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
merchant-portal-prod 386906790760 · Root / Workloads / Production	$7.0k	Moderate	▾ Moderate 4 **Moderate** **RDS / Aurora** $2.5k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EC2** $2.1k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **ECS** $1.4k/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. **Moderate** **Lambda** $1.1k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.

Account	Spend / month	Readiness	Multi-region risks
disputes-prod 844781801768 · Root / Workloads / Production	$6.9k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $3.5k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $2.1k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $685/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $681/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
card-controls-prod 170174047247 · Root / Workloads / Production	$6.4k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $3.2k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.9k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $642/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $638/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
pricing-prod 322790187289 · Root / Workloads / Production	$6.1k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $3.0k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.8k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $613/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $609/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
collections-prod 110356959844 · Root / Workloads / Production	$6.0k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $3.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $1.2k/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $1.2k/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
audit-trail-prod 252831874844 · Root / Workloads / Production	$5.4k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $2.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $533/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $537/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
loans-origination-prod 369887811678 · Root / Workloads / Production	$5.3k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $2.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.6k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $529/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $530/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
sms-gateway-prod 698441060454 · Root / Workloads / Production	$4.7k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $2.4k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.4k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $470/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $471/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
kyc-checks-prod 706271427442 · Root / Workloads / Production	$4.5k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $2.3k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.3k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $450/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $453/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
atm-switch-prod 268805426379 · Root / Workloads / Production	$4.5k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $2.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $896/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $893/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
legacy-batch-prod 107537456074 · Root / Workloads / Production	$4.2k	High	▾ High 1 · Moderate 2 · Low 1 **High** **Data Pipeline** $327/mo · No live replica: restore or re-run Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours. **Moderate** **EC2** $2.6k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **SWF** $712/mo · Manual or scripted failover Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. **Low** **S3** $536/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
search-prod 899894212661 · Root / Workloads / Production	$4.0k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **Lambda** $1.4k/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **API Gateway** $793/mo · Manual or scripted failover Deploy the API in both regions behind a custom domain with Route 53 failover. **Moderate** **SQS** $396/mo · Manual or scripted failover Queues are regional and messages are not replicated: in-flight messages are lost or delayed on failover. **Low** **DynamoDB** $1.4k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.

Account	Spend / month	Readiness	Multi-region risks
geo-services-prod 154224630493 · Root / Workloads / Production	$4.0k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $2.0k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.2k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $401/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $397/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
chatbot-prod 925052687979 · Root / Workloads / Production	$3.4k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $1.7k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $1.0k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $343/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $342/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
statements-prod 672004302296 · Root / Workloads / Production	$2.9k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $1.0k/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $881/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $591/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $438/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.

Account	Spend / month	Readiness	Multi-region risks
treasury-reporting-prod 499647302627 · Root / Workloads / Production	$2.8k	Moderate	▾ Moderate 3 · Low 2 **Moderate** **QuickSight** $781/mo · Manual or scripted failover Has its own failover model, often only for specific region pairs. **Moderate** **Lambda** $533/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. **Moderate** **Step Functions** $287/mo · Manual or scripted failover State machines are regional; running executions do not fail over and must be restarted. **Low** **DynamoDB** $1.0k/mo · AWS replicates and fails over Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. **Low** **S3** $223/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
notifications-prod 117309034122 · Root / Workloads / Production	$2.6k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $1.3k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $782/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $260/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $263/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
partner-api-prod 283827598105 · Root / Workloads / Production	$2.2k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $1.1k/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $664/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $222/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $223/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
onboarding-prod 149964263026 · Root / Workloads / Production	$2.1k	Moderate	▾ Moderate 4 Moderate **RDS / Aurora** $722/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. Moderate **EC2** $627/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. Moderate **ECS** $414/mo · Manual or scripted failover Clusters and services are regional: deploy to both regions and fail over with Route 53. Moderate **Lambda** $315/mo · Manual or scripted failover Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.

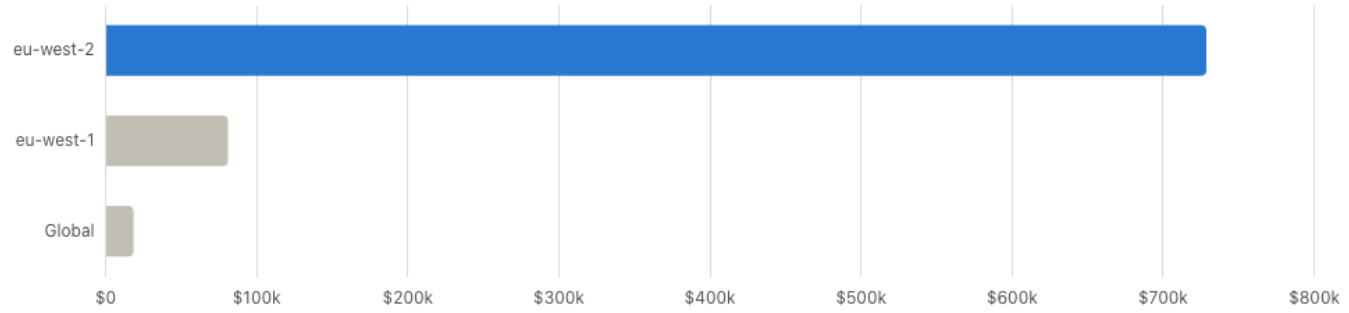
Account	Spend / month	Readiness	Multi-region risks
rewards-prod 271987579183 · Root / Workloads / Production	$1.7k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $829/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $497/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $166/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $164/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.
limits-prod 152477260104 · Root / Workloads / Production	$1.4k	Moderate	▾ Moderate 2 · Low 1 **Moderate** **EC2** $867/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **EC2 Other (EBS, NAT, transfer)** $292/mo · Manual or scripted failover EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region. **Low** **S3** $289/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

Account	Spend / month	Readiness	Multi-region risks
fx-rates-prod 753954983820 · Root / Workloads / Production	$1.0k	Moderate	▾ Moderate 3 · Low 1 **Moderate** **EC2** $518/mo · Manual or scripted failover Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations. **Moderate** **RDS / Aurora** $313/mo · Manual or scripted failover Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost. **Moderate** **EKS** $105/mo · Manual or scripted failover A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication. **Low** **S3** $104/mo · AWS replicates and fails over Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.

4 Which second region

Where the chosen accounts run today

Monthly spend by region.



Recommended: Europe (Frankfurt)

Best balance of service parity, latency and price for the chosen accounts and recovery target.

Region	Europe (Frankfurt) (eu-central-1)
Estimated latency from primary	~15 ms round trip
Service parity (by spend)	100.0%
Not available there	None
Prices against primary	+2%
Already runs there	0.0% of regional spend

Candidate regions ranked by service parity, latency and price

Region	Latency	Parity	Not available	Price	Score	Why it ranks here
Europe (Frankfurt) eu-central-1 **Recommended**	~15 ms	100.0%	none	+2%	87	Best balance of service parity, latency and price.
Europe (Ireland) eu-west-1	~10 ms	97.4%	Connect	-4%	86	No Connect here ($20.9k/month to re-architect). 10% of spend already runs here.
US East (N. Virginia) us-east-1	~95 ms	100.0%	none	-14%	84	~80 ms further than Europe (Frankfurt).
Canada (Central) ca-central-1	~85 ms	100.0%	none	-5%	83	~70 ms further than Europe (Frankfurt).
US West (Oregon) us-west-2	~120 ms	100.0%	none	-14%	82	~105 ms further than Europe (Frankfurt).
Europe (Stockholm) eu-north-1	~25 ms	96.7%	Connect, Lex, Textract	-9%	79	No Connect, Lex, Textract here ($26.9k/month to re-architect); ~10 ms further than Europe (Frankfurt).
Europe (Paris) eu-west-3	~10 ms	97.0%	Connect, Lex	+1%	78	No Connect, Lex here ($24.0k/month to re-architect).
Europe (Spain) eu-south-2	~20 ms	96.4%	Amplify, Connect, Lex, Transcribe	-3%	77	No Amplify, Connect, Lex and others here ($28.7k/month to re-architect).
Asia Pacific (Seoul) ap-northeast-2	~140 ms	100.0%	none	+5%	76	~125 ms further than Europe (Frankfurt); about 3% dearer than Europe (Frankfurt).
US East (Ohio) us-east-2	~100 ms	97.0%	Connect, Lex	-14%	75	No Connect, Lex here ($24.0k/month to re-architect); ~85 ms further than Europe (Frankfurt).

Regulation to check before data moves to Europe (Frankfurt); pointers, not legal advice

UK GDPR (international transfers)

Outside the UK, a transfer to Germany needs UK adequacy regulations or safeguards (the International Data Transfer Agreement or the UK Addendum to the EU clauses).

PRA SS2/21 (outsourcing and third-party risk management)

Notify the PRA of material outsourcing changes, which include where data is held, and include the second region in exit and stressed-exit plans.

5 Cost to run, and the people

Second-region infrastructure

$2m to $4.5m PROVISIONAL

a year, incl. replication traffic

AWS Support uplift

$60k to $140k PROVISIONAL

Enterprise plan, a year

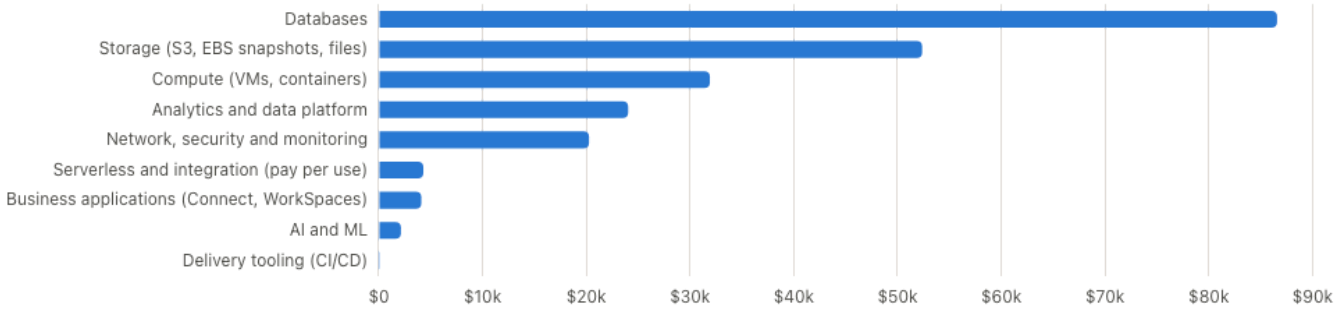
Extra people

$400k to $800k PROVISIONAL

2 to 5 FTE a year

Second-region infrastructure by service class

Monthly, after the recovery target's share and regional prices.



Extra people to run two regions

Role	FTE (midpoint)
Platform and SRE: on-call and operations for two regions	1.0
Resilience: DR testing, failover rehearsals, regulator evidence	1.0
Workload operations across multi-region accounts	0.4
Data replication and database operations	0.8
Total (quoted as 2 to 5)	**3.2**

At $165k per FTE a year.

6 Cost and time to achieve

Cost to achieve (one-off)

$1.6m to $3.5m PROVISIONAL

engineering, plus running the region while it is built

Time to achieve

5 to 11 months PROVISIONAL

bound by team capacity · team of 15 full-time people

Engineering effort

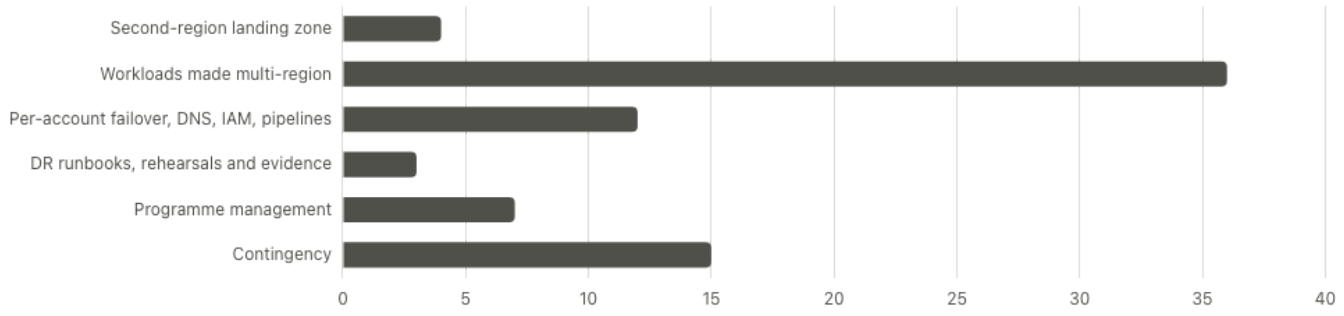
4.5 to 10 person-years

PROVISIONAL

40 material accounts

Where the effort goes

Person-months, after the delivery-maturity adjustment.



7 Service parity in the second region

Service	Class	Accounts	Spend / month	In eu-central-1
EC2	Compute (VMs, containers)	33	$188k	Available
RDS / Aurora	Databases	25	$107k	Available
Redshift	Analytics and data platform	1	$62k	Available
S3	Storage (S3, EBS snapshots, files)	31	$55k	Available
DynamoDB	Databases	6	$44k	Available
EC2 Other (EBS, NAT, transfer)	Compute (VMs, containers)	9	$43k	Available
Lambda	Serverless and integration (pay per use)	10	$33k	Available
SageMaker	AI and ML	2	$23k	Available
Connect	Business applications (Connect, WorkSpaces)	1	$21k	Available
Glue	Analytics and data platform	1	$15k	Available
ECS	Compute (VMs, containers)	5	$14k	Available
EKS	Compute (VMs, containers)	19	$14k	Available
VPC	Network, security and monitoring	1	$12k	Available
Direct Connect	Network, security and monitoring	2	$11k	Available
API Gateway	Serverless and integration (pay per use)	2	$11k	Available
EMR	Analytics and data platform	1	$11k	Available
Kinesis Streams	Databases	2	$9.7k	Available
CloudWatch	Network, security and monitoring	2	$8.4k	Available
Cognito	Serverless and integration (pay per use)	2	$8.0k	Available
MSK (Kafka)	Databases	1	$7.9k	Available
Bedrock	AI and ML	1	$7.6k	Available
Step Functions	Serverless and integration (pay per use)	2	$7.4k	Available
Neptune	Databases	1	$6.8k	Available

Service	Class	Accounts	Spend / month	In eu-central-1
Athena	Serverless and integration (pay per use)	1	$6.6k	Available
QuickSight	Business applications (Connect, WorkSpaces)	2	$5.9k	Available
FSx	Storage (S3, EBS snapshots, files)	1	$5.7k	Available
Network Firewall	Network, security and monitoring	1	$5.3k	Available
OpenSearch	Databases	1	$4.7k	Available
AppSync	Serverless and integration (pay per use)	1	$4.3k	Available
ELB	Network, security and monitoring	2	$4.2k	Available
ElastiCache	Databases	1	$4.2k	Available
Managed Flink	Analytics and data platform	1	$3.6k	Available
SQS	Serverless and integration (pay per use)	2	$3.6k	Available
Transcribe	AI and ML	1	$3.4k	Available
Backup	Storage (S3, EBS snapshots, files)	1	$3.4k	Available
Lex	AI and ML	1	$3.1k	Available
Textract	AI and ML	1	$3.0k	Available
Pinpoint	Serverless and integration (pay per use)	1	$2.8k	Available
Firehose	Serverless and integration (pay per use)	1	$2.6k	Available
KMS	Network, security and monitoring	2	$2.4k	Available
Directory Service	Network, security and monitoring	1	$2.2k	Available
MWAA (Airflow)	Analytics and data platform	1	$2.0k	Available
Timestream	Databases	1	$2.0k	Not verified
SNS	Serverless and integration (pay per use)	1	$1.8k	Available
EventBridge	Serverless and integration (pay per use)	1	$1.7k	Available

Service	Class	Accounts	Spend / month	In eu-central-1
ACM / Private CA	Network, security and monitoring	1	$1.6k	Available
WAF	Network, security and monitoring	1	$1.5k	Available
Amazon MQ	Databases	1	$1.4k	Available
Amplify	Serverless and integration (pay per use)	1	$1.3k	Available
Bedrock AgentCore	AI and ML	1	$1.2k	Available
Transform	Compute (VMs, containers)	1	$495	Not verified
Data Pipeline	Analytics and data platform	1	$327	Not verified

Assumptions

- Recovery target: Recover within 3 hours of a region-wide event. Each service's spend in the chosen accounts is multiplied by its class's second-region share (data/regions.yaml): Compute (VMs, containers) 12%; Databases 45%; Storage (S3, EBS snapshots, files) 80%; Analytics and data platform 25%; Network, security and monitoring 40%; Serverless and integration (pay per use) 5%; Business applications (Connect, WorkSpaces) 15%; AI and ML 5%; Delivery tooling (CI/CD) 5%.
- Replication traffic: 6% of database, storage and analytics spend.
- Second-region prices use an indicative regional price index (EC2 on-demand list prices against us-east-1).
- AWS Support: Enterprise list-price tiers applied to the whole scanned bill plus the new region; enterprise agreements and discounts differ.
- Latency is estimated from distance (fibre ~200 km/ms, routes 1.5x the great circle): measure before committing.
- Service availability: AWS's public regional services table, version 20251113063700 (fetched 2026-10-21), corrected where it is known to be wrong (Amazon Lex available in eu-west-1). Services not listed there are 'not verified' and their effort is raised 25%.
- Region score: parity 55%, latency 20%, price 25%, plus a bonus where the estate already runs.
- People: 10% of the core cloud team (10) + 1 resilience + 0.25 per 25 material accounts + data operations scaled by data spend; at least 1.5. Cost per FTE 165,000 USD a year.
- Build: catalog-style engineer-weeks per service and account × delivery factor 1.26 (CI/CD × IaC × regulated), plus a second-region landing zone (halved where 5%+ of spend already runs there), DR programme, 12% programme management and 25% contingency.
- Duration: the longer of effort ÷ capacity (50% of the cloud team + application engineers, 80% productive) and a 20-week critical path, plus 4 weeks of failover rehearsal. Build cost includes running the half-built region meanwhile.
- Figures are ranges (−25% / +50% on cost, −20% / +40% on time), widened to round numbers.